



Impact of Virtual Agents' Behaviour on Obedience and Social Norm Compliance

Group number: CP1_a

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Abstract: This experiment explored how the interpersonal attitude of virtual agents affects people's obedience and compliance with social norms. Participants were tasked with retrieving a coffee cup and could choose to walk behind the agents or through their o-space. The experiment measured the waiting time from when the agents gave the waiting instruction to task completion, as well as participants' route choices. Results showed that participants were more likely to invade the agents' o-space when there was no interaction, while this tendency decreased when the agents left a friendlier impression. Interestingly, the agents' instructions had little impact on waiting time.

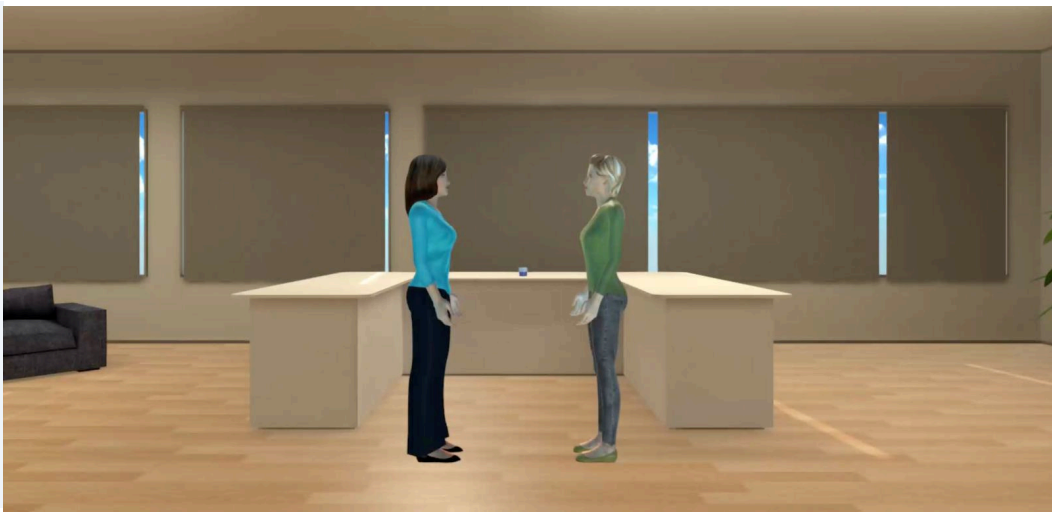
Background

This study was based on Pascoe's (2024) and Zojaji's (2020) research, who previously have shown that when virtual agents are more friendly, people tend to obey their command and to adhere to social norms by walking around them. Pascoe's (2024) demonstrated that people broke the o-space most often in conditions where there was no interaction from agents. The research of Zalake et al. (2021) and Vinayagamoorthy et al. (2006) also supports that people are more likely to follow instructions from agents who are friendlier.

Method

Six participants watched videos of agents and verbally indicated their paths to the coffee cup. Eight participants completed the task in VR, physically walking to collect the cup after agents instructed them to wait.

Each participant performed the task under six conditions: Friendly-Dominant, Friendly-Submissive, Hostile-Dominant, Neutral-Non-Verbal, and Control. Measurements included task time, path taken, agent attitude, and VR immersion. Equipment used: Meta Quest 2 and an HP Pavilion computer.



Results

How often participants walked around the agents for each condition.

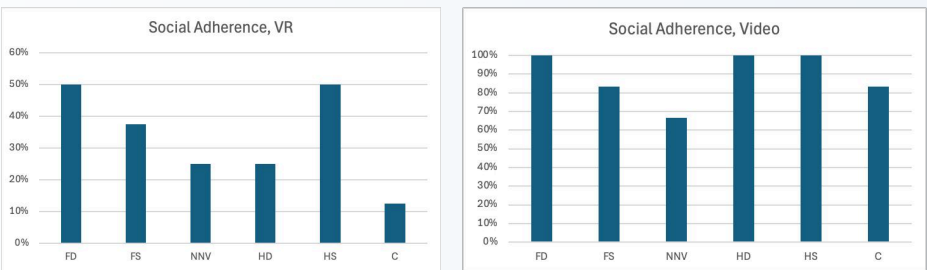


Figure 1

Figure 2

How much the participants like the agents.

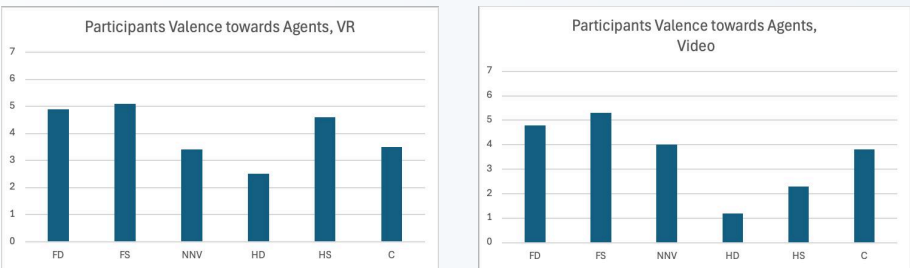


Figure 3

Figure 4

Time taken to complete the task was used to calculate their waiting time, which indicates their obedience towards the agents.

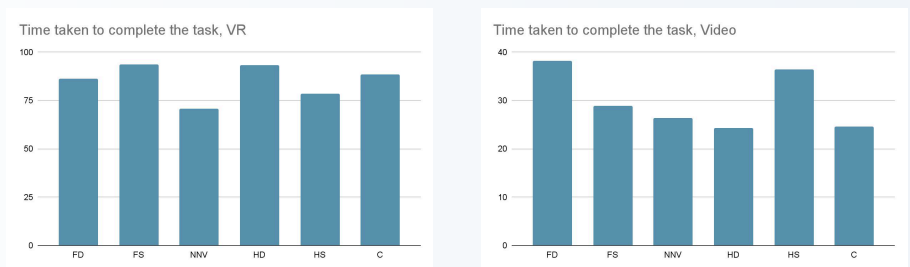


Figure 5

Figure 6



Conclusion

Our data and interviews showed that the interpersonal attitude of social agents influenced participants' behavior in both virtual and video-based experiments. Unexpectedly, there was no significant increase in waiting time when participants were explicitly asked to wait.

Friendlier agents led to greater adherence to social norms and less frequent crossing of the o-space, highlighting the role of agent HCI design in influencing user behavior.

References

Pascoe, E. J. D. (2024). Effect of a dyad of agents' interpersonal attitude on obedience and social norm adherence in a virtual reality scenario (Degree project, Second cycle, 30 credits). KTH Royal Institute of Technology.

Vinayagamoorthy, V., Gillies, M., Steed, A., Tanguy, E., Pan, X., Loscos, C., & Slater, M. (2006). Building expression into virtual characters. In Eurographics 2006. Vienna, Austria: Eurographics Association.

Zalake, M., Siqueira, A. G. de, Vaddiparti, K., & Lok, B. (2021). The effects of virtual human's verbal persuasion strategies on user intention and behavior. International Journal of Human-Computer Studies, 156, 102708. <https://doi.org/10.1016/j.ijhcs.2021.102708>

Zojaji, S., Peters, C., & Pelachaud, C. (2020). Influence of virtual agent politeness behaviors on how users join small conversational groups. Proceedings of the 20th ACM International Conference on Intelligent Virtual Agents, IVA 2020. Presented at the 20th ACM International Conference on Intelligent Virtual Agents, IVA '20, Virtual Event, Scotland, UK, October 20-22, 2020. <https://doi.org/10.1145/3383652.3423917>